GOWTHAM D P

Mechanical Engineering Fresher | PLC Automation | CAD Design | Embedded Systems

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SUMMARY

Mechanical Engineering student with hands-on experience in Fusion 360, AutoCAD, and PLC-based automation systems. Skilled in 3D modeling, prototype development, and Arduino-based embedded projects with strong attention to manufacturable design principles. Seeking opportunities in product design and engineering development roles where I can contribute practical CAD solutions and innovative product concepts.

PROFESSIONAL EXPERIENCE

MIPZO EdTech Pvt Ltd — **DEC 2024 - APR 2025**
Research and Development Intern

- Designed and developed engineering prototypes using Fusion 360 and AutoCAD for robotics and IoT learning applications
- Integrated Arduino-based sensor modules including ultrasonic and IR sensors for embedded system experiments
- Performed functional testing and debugging of prototype circuits to improve reliability and performance
- Supported component selection and optimized material usage during prototype development

CAD Design Experience

- Created parametric 3D part models using Autodesk Fusion 360 for mechanical and automation applications
- Developed multi-component assemblies for embedded and sensor-based prototype systems
- Generated 2D technical drawings with dimensions for fabrication reference
- Optimized component layouts to improve compactness, functionality, and assembly feasibility

EDUCATION

Government Engineering College, Hassan — **Sept 2025 - Present**
Bachelor of Mechanical Engineering

Smt L V (Government) Polytechnic, Hassan — **Oct 2022 - Apr 2025**
Diploma in Mechanical Engineering - **9.24 CGPA**

Government Primary and Secondary School — **Jun 2012 - Apr 2022**
KBSE - **89.06%**

PROJECT

Mar 2023 - Apr 2023

Obstacle Avoiding Sensor Car

- Designed and developed an Arduino-based autonomous robotic vehicle using ultrasonic sensors and L298N motor driver
- Implemented real-time obstacle detection and navigation control logic using embedded system programming
- Integrated sensor modules and motor driver circuitry on a custom chassis platform for autonomous movement testing

PLC-Based Water Level Control System **Mar 2024 - Apr 2025**

- Designed and implemented a PLC-based automated water level control system using level sensors
- Programmed ladder logic to control pump ON/OFF operation and prevent dry-run and overflow conditions
- Integrated sensor inputs with PLC control logic to improve system reliability and reduce manual monitoring effort
- Tested system performance under different tank level conditions during implementation

Dual-Tank Automated Drainage Control System using PLC **Sep 2024 - Oct 2024**

- Designed and implemented a dual-tank automated drainage control system using PLC ladder logic
- Controlled tank filling and draining operations based on central storage level conditions using level sensors
- Integrated sensor inputs with PLC control logic to automate sequential tank operation and reduce manual monitoring
- Improved process efficiency and reliability through automated tank-level management logic

TECHNICAL SKILLS

Programming Languages

Python,HTML,Css

Technologies/Frameworks

CAD Design Software:

- AutoCAD
- Autodesk Fusion 360
- Autodesk Inventor
- Parametric 3D Part Modeling
- Parametric 3D Assembly Modeling
- 2D Manufacturing Drawing Creation

Simulation Tools:

- ANSYS (Student Version)

Industrial Automation

- Programmable Logic Controller (PLC)
- PLC Ladder Logic
- HMI Basics
- SCADA Basics
- Sensor-Based Automation Systems

EMBEDDED SYSTEMS

- Arduino UNO
- ESP Modules
- Ultrasonic Sensor Integration
- IR Sensor Integration
- Arduino IDE

Languages

- English – Professional Working Proficiency
- Kannada – Native Proficiency
- Hindi – Basic Proficiency

Certifications

- PLC, HMI & SCADA Training – DesignTech & GT&TC
- PLC & Industrial Automation – Instrumentation Tools
- R&D Internship (Robotics & IoT) – MIPZO EdTech Pvt. Ltd.
- RoboSoccer Workshop (12 Days) – Malnad College of Engineering (MCE)

Technical Interests

- Industrial Automation
- Embedded Systems
- Product Design using CAD
- IoT & Robotics
- Open-Source Engineering Projects